

GOWTHAM C

cgowtham193@gmail.com | 9003480578 | Madurai, Tamil Nadu | LinkedIn | Portfolio

CAREER OBJECTIVE

Self-motivated and enthusiastic Software Developer with a solid foundation in Python and C. Passionate computer application graduate seeking an entry-level software engineering position to leverage programming skills and contribute to innovative technology solutions

PROJECTS

Tamper-Resistant Cryptographic

Authentication System

- **Technical Stack:** Python, C++, ESP32, PySerial, PyAutoGUI, NVS Flash.
- **Core Innovation:** Engineered an "Air-Gapped" security model where credentials are stored in the ESP32's **Non-Volatile Storage (NVS)**, invisible to the host PC's file system.
- **Context-Aware Logic:** Built a background Python service to monitor active window titles and automatically "prime" the hardware when login portals (e.g., GitHub, AWS) are detected.
- **Security Hardening:** Utilized **HID (Human Interface Device) emulation** to inject passwords, bypassing software-level keyloggers and the system clipboard.

Cloud Cost Monitoring and Optimization Tool,

A Flask and AWS-Based Cost Monitoring System

- Developed a cloud financial management tool using **Python (Boto3)** and **Flask** to track real-time **AWS** resource usage.
- Implemented cost-estimation logic to identify idle instances and underutilized services, helping in resource optimization.

Conventional Planner Festive Frolics

- A web-based platform designed for hassle-free event management by providing a centralized booking system.
- The system improves coordination and saves time by allowing users to plan and book multiple event services in one place.

EDUCATION

Master of Computer Applications,

Rathinam Technial Campus Coimbatore

2024 – 2026

Bachelor of Computer Applications,

The American College Madurai

2021 – 2024

SKILLS

- Programming Languages: C, C++ (Core), Python (Boto3, Flask), Java (Basic)
- OOP & Design: Object-Oriented Analysis and Design (OOAD), Data Structures, Design Patterns
- Frontend Technologies: HTML, CSS, JavaScript
- Frameworks & Tools: Flask, .NET Framework (Basic), C# (Basic), Qt (Basic GUI concepts)
- Development Practices: Unit Testing, Code Review, Software Development Lifecycle (SDLC), Process Compliance
- Tools & IDEs: Visual Studio Code, Code Blocks, Vim Editor, ESP32 Microcontroller

SOFT SKILLS

- Soft Skills: Logical Thinking, Team Collaboration, Quick Learner, Accountability, Problem-Solving
- Interests: Hosting Podcasts, Reading Books, Learning new Software Industry trends

CERTIFICATES

- **C and C++ Programming:** Certified by SSI Technology.
- Preparing for IEEE-style paper publication on **Hardware Security**.